

FUNDAMENTAL THEOREM OF ISOMORPHISM IN GROUP THEORY

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Declaration

I hereby declare that I am the sole author of this Reading Project Report in fulfillment of the requirements for Summer Internship from National Institute of Science Education and Research (NISER).

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The Project Report work entitled "Fundamental Theorem of Isomorphism in Group Theory" was carried out under my supervision, in the School of Mathematical Sciences at NISER, Bhubaneswar, India.

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Preface and Acknowledgement

This project report documents the work done during the summer internship at the National Institute of Science Education and Research (NISER), Bhubaneswar, India, from May 13, 2024, to July 9, 2024, focusing on the Fundamental Theorem of Homomorphisms in Group Theory under the mentorship of Dr. Binod Kumar Sahoo. The report provides an overview of the tasks accomplished during this internship period. $\text{\LaTeX}$ was used to edit this project report, aiming to maintain simplicity alongside technical accuracy.

I express my sincere gratitude to NISER Bhubaneswar for providing the necessary facilities and environment in completing my reading project smoothly. I am deeply thankful to Dr. Binod Kumar Sahoo, School of Mathematical Sciences, NISER Bhubaneswar, for dedicating his valuable time to supervise and guide me throughout the journey. His unwavering support, expert guidance, and constant encouragement were instrumental in shaping this project.

I would also like to extend my heartfelt thanks to my parents, siblings, and close friends for their unwavering support and motivation in every aspect of my life.

Abstract

This report provides an extensive exploration of the Fundamental Theorem of Homomorphisms in group theory, a fundamental concept with profound implications across mathematics. Beginning with the foundational definitions and properties of groups, including illustrative examples to solidify understanding, we establish the groundwork necessary for deeper exploration. Subsequent sections delve into subgroups, emphasizing cyclic groups and their significance in the context of group theory. The concept of cosets is introduced, followed by an examination of normal subgroups and quotient groups, elucidating their roles in group structure and manipulation.

Central to this report is the comprehensive discussion on homomorphisms—functions that preserve the structure of groups—beginning with their definition and extending to their diverse applications. The Fundamental Theorem of Homomorphisms is the focal point, offering a profound insight into the relationship between homomorphisms, their kernels, and the quotient groups they induce. Through detailed proofs, examples, and practical applications, the theorem's theoretical underpinnings and practical utility are thoroughly explored.

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Chapter 1

Introduction to Groups

Groups are fundamental structures in abstract algebra that lay the groundwork for studying symmetry, transformations, and algebraic operations. They consist of a set of elements and a defined way to combine them, satisfying rules like closure (the result stays within the set), associativity (order of operations doesn't matter), identity (an element that doesn't change others), and invertibility (each element has an opposite).

Groups are essential building blocks alongside rings, fields, and vector spaces. Studying groups introduces important algebraic concepts like homomorphisms (structure-preserving mappings) and quotient constructions (forming new groups from existing ones), which are crucial in abstract algebra and have wide applications in mathematics and beyond.

The focus on groups in abstract algebra comes from their simplicity and broad usefulness. Starting with groups provides a strong foundation for exploring more complex algebraic structures.

Historically, groups were first studied in specific cases, such as permutations and symmetries, before being formally defined in the late 19th century. This abstract approach allows mathematicians to apply these concepts broadly, deepening our understanding of algebraic systems and supporting advances in both theoretical mathematics and practical applications.

1.1 Definition of a Group

Definition: A nonempty set of elements G is said to form a *group* if in G there is defined a binary operation, called the product and denoted by $\cdot$, such that

1. Closure: $a, b \in G$ implies that $a \cdot b \in G$.
2. Associative: $a, b, c \in G$ implies that $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
3. Existence of an identity element in G : There exist an element $e \in G$ such that $a \cdot e = e \cdot a = a$ for all $a \in G$.
4. Existence of inverses in G : For every $a \in G$ there exist an element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

Definition: A group G is said to be *abelian* or *commutative* if for every $a, b \in G$, $a \cdot b = b \cdot a$. A group which is not abelian is called *non-abelian*.

Order of a group: The order of a finite group is its cardinality, i.e., the number of elements present in group G . It is denoted by $o(G)$.

This number plays a key role in subgroup classification, group actions, and theoretical explorations within abstract algebra.

1.2 Some Examples of Groups

1. The set of integers $\mathbb{Z}$ under addition, denoted as $(\mathbb{Z}, +)$, is a group.

Closure: For any integers $a, b \in \mathbb{Z}$, $a + b$ is also an integer.

Associativity: For all integers $a, b, c \in \mathbb{Z}$, $a + (b + c) = (a + b) + c$.

Identity Element: 0 is the identity element, as $a + 0 = 0 + a = a$ for all $a \in \mathbb{Z}$.

Inverse Element: For each $a \in \mathbb{Z}$, there exists an element $-a \in \mathbb{Z}$ such that $a + (-a) = (-a) + a = 0$.

Therefore, $(\mathbb{Z}, +)$ forms a group under addition.

2. The set of integers modulo n , denoted as $\mathbb{Z}_n$, forms a group under addition modulo n .

Closure: For any $a, b \in \mathbb{Z}_n$, $a +_n b$ (which is $(a + b) \bmod n$) is also in $\mathbb{Z}_n$.

Associativity: Addition modulo n is associative: $(a +_n b) +_n c = a +_n (b +_n c)$ for all $a, b, c \in \mathbb{Z}_n$.

Identity Element: The identity element in $\mathbb{Z}_n$ under addition modulo n is 0: $a +_n 0 = 0 +_n a = a$ for all $a \in \mathbb{Z}_n$.

Inverse Element: Every element $a \in \mathbb{Z}_n$ has an inverse under addition modulo n , which is $n - a$: $a +_n (n - a) \equiv 0 \bmod n$.

Therefore, $(\mathbb{Z}_n, +_n)$ forms a group under addition modulo n .

1.3 Some Preliminary Lemmas

Lemma 1. *If G is a group, then*

1. *The identity element of G is unique.*
2. *Every $a \in G$ has a unique inverse in G .*
3. *For every $a \in G$, $(a^{-1})^{-1} = a$.*
4. *For all $a, b \in G$, $(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$.*

Proof.

1. Let e and e' be both identity elements in G . By the definition of an identity element, we have:

$$e \cdot e' = e \quad \text{and} \quad e' \cdot e = e'.$$

Therefore, $e = e'$. Hence, the identity element in G is unique.

2. Let $a \in G$. Suppose b and c are both inverses of a in G . Then, by definition of inverses, we have:

$$a \cdot b = e \quad \text{and} \quad a \cdot c = e,$$

where e is the identity element in G . Multiplying both sides of $a \cdot b = e$ on the right by c , we get:

$$a \cdot b \cdot c = e \cdot c = c.$$

Since $a \cdot c = e$, we have:

$$a \cdot b \cdot c = e \cdot c = c = e \cdot b = b.$$

Therefore, $b = c$, proving that the inverse of a is unique.

3. Let $a \in G$. By definition, a^{-1} is the inverse of a , so:

$$a \cdot a^{-1} = e \quad \text{and} \quad a^{-1} \cdot a = e,$$

where e is the identity element in G . Let $b = a^{-1}$. Then, by the same definition, b^{-1} is the inverse of b , so:

$$b \cdot b^{-1} = e \quad \text{and} \quad b^{-1} \cdot b = e.$$

Substituting $b = a^{-1}$, we get:

$$a^{-1} \cdot (a^{-1})^{-1} = e \quad \text{and} \quad (a^{-1})^{-1} \cdot a^{-1} = e.$$

Thus, $(a^{-1})^{-1} = a$.

4. Let $a, b \in G$. We need to show that $(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$. We have:

$$(a \cdot b) \cdot (b^{-1} \cdot a^{-1}) = a \cdot (b \cdot b^{-1}) \cdot a^{-1} = a \cdot e \cdot a^{-1} = a \cdot a^{-1} = e,$$

and

$$(b^{-1} \cdot a^{-1}) \cdot (a \cdot b) = b^{-1} \cdot (a^{-1} \cdot a) \cdot b = b^{-1} \cdot e \cdot b = b^{-1} \cdot b = e.$$

Therefore, $(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$.

Lemma 2. *Given a, b in the group G , then the equations $a \cdot x = b$ and $y \cdot a = b$ have unique solutions for x and y in G . In particular, the two cancellation laws hold in G :*

$$1. \quad a \cdot u = a \cdot w \implies u = w,$$

$$2. \quad u \cdot a = w \cdot a \implies u = w.$$

Proof. Let $a, b \in G$ and consider the equation $a \cdot x = b$.

Since G is a group, every element $a \in G$ has an inverse $a^{-1} \in G$ such that $a \cdot a^{-1} = e$ and $a^{-1} \cdot a = e$, where e is the identity element in G .

To solve $a \cdot x = b$ for x , multiply both sides on the left by a^{-1} :

$$a^{-1} \cdot (a \cdot x) = a^{-1} \cdot b.$$

Using the associativity property of the group operation, we get:

$$(a^{-1} \cdot a) \cdot x = a^{-1} \cdot b.$$

Since $a^{-1} \cdot a = e$, we have:

$$e \cdot x = a^{-1} \cdot b,$$

and since $e \cdot x = x$, it follows that:

$$x = a^{-1} \cdot b.$$

Thus, the equation $a \cdot x = b$ has a unique solution $x = a^{-1} \cdot b$.

Similarly, consider the equation $y \cdot a = b$. To solve for y , multiply both sides on the right by a^{-1} :

$$(y \cdot a) \cdot a^{-1} = b \cdot a^{-1}.$$

Using associativity, we get:

$$y \cdot (a \cdot a^{-1}) = b \cdot a^{-1}.$$

Since $a \cdot a^{-1} = e$, we have:

$$y \cdot e = b \cdot a^{-1},$$

and since $y \cdot e = y$, it follows that:

$$y = b \cdot a^{-1}.$$

Thus, the equation $y \cdot a = b$ has a unique solution $y = b \cdot a^{-1}$.

To prove the cancellation laws, let $a \cdot u = a \cdot w$. Multiplying both sides on the left by a^{-1} , we get:

$$a^{-1} \cdot (a \cdot u) = a^{-1} \cdot (a \cdot w).$$

Using associativity and the property $a^{-1} \cdot a = e$, we obtain:

$$(e \cdot u) = (e \cdot w),$$

which simplifies to:

$$u = w.$$

For the second cancellation law, let $u \cdot a = w \cdot a$. Multiplying both sides on the right by a^{-1} , we get:

$$(u \cdot a) \cdot a^{-1} = (w \cdot a) \cdot a^{-1}.$$

Using associativity and the property $a \cdot a^{-1} = e$, we obtain:

$$u \cdot (a \cdot a^{-1}) = w \cdot (a \cdot a^{-1}),$$

which simplifies to:

$$u \cdot e = w \cdot e,$$

and since $u \cdot e = u$ and $w \cdot e = w$, it follows that:

$$u = w.$$

Thus, both cancellation laws hold in G .

1.4 Few Examples

1. Consider the set $G = \{1, a, a^2, a^3\}$, where a is an element such that $a^4 = 1$ (the identity element). Define the operation as multiplication modulo 4. Determine whether G forms a group under this

operation. If it does, verify whether G is abelian.

Closure: To verify closure under multiplication modulo 4:

$$\begin{aligned}
1 \cdot 1 &\equiv 1 \pmod{4}, \\
1 \cdot a &\equiv a \pmod{4}, \\
1 \cdot a^2 &\equiv a^2 \pmod{4}, \\
1 \cdot a^3 &\equiv a^3 \pmod{4}, \\
a \cdot a &\equiv a^2 \pmod{4}, \\
a \cdot a^2 &\equiv a^3 \pmod{4}, \\
a \cdot a^3 &\equiv 1 \pmod{4}, \\
a^2 \cdot a^2 &\equiv a^3 \pmod{4}, \\
a^2 \cdot a^3 &\equiv 1 \pmod{4}, \\
a^3 \cdot a^3 &\equiv a \pmod{4}.
\end{aligned}$$

Since each product $g \cdot h \pmod{4}$ is in G , G is closed under the defined operation.

Identity Element: The element 1 serves as the identity:

$$1 \cdot g \equiv g \cdot 1 \equiv g \pmod{4} \quad \text{for all } g \in G.$$

Inverses:

$$\begin{aligned}
1^{-1} &= 1, \quad a^{-1} = a^3 \text{ (since } a \cdot a^3 \equiv 1 \pmod{4}), \\
(a^2)^{-1} &= a^2 \text{ (since } a^2 \cdot a^2 \equiv 1 \pmod{4}), \\
(a^3)^{-1} &= a \text{ (since } a^3 \cdot a \equiv 1 \pmod{4}).
\end{aligned}$$

Associativity: Multiplication modulo 4 is associative by definition of modular arithmetic.

Abelianness:

$$\begin{aligned}
1 \cdot a &= a \cdot 1, \quad 1 \cdot a^2 = a^2 \cdot 1, \quad 1 \cdot a^3 = a^3 \cdot 1, \\
a \cdot a^2 &= a^2 \cdot a, \quad a \cdot a^3 = a^3 \cdot a, \quad a^2 \cdot a^3 = a^3 \cdot a^2.
\end{aligned}$$

Since all products commute, G is abelian.

Therefore, $G = \{1, a, a^2, a^3\}$ forms a group under multiplication modulo 4 and is abelian.

2. Prove that if G is an abelian group, then for all $a, b \in G$ and all integers n ,

$$(a \cdot b)^n = a^n \cdot b^n.$$

Proof. Assume that G is an abelian group. We aim to prove that for all $a, b \in G$ and all integers n ,

$$(a \cdot b)^n = a^n \cdot b^n.$$

For $n = 2$:

$$(a \cdot b)^2 = (a \cdot b)(a \cdot b) = a \cdot b \cdot a \cdot b = a \cdot a \cdot b \cdot b = a^2 \cdot b^2.$$

Let $(a \cdot b)^{n-1} = a^{n-1} \cdot b^{n-1}$.

We prove that $(a \cdot b)^n = a^n \cdot b^n$:

$$(a \cdot b)^n = (a \cdot b)^{n-1} \cdot (a \cdot b) = a^{n-1} \cdot b^{n-1} \cdot (a \cdot b) = a^{n-1} \cdot a \cdot b^{n-1} \cdot b = a^n \cdot b^n.$$

Therefore, if G is an abelian group, for all $a, b \in G$ and all integers n ,

$$(a \cdot b)^n = a^n \cdot b^n.$$

3. If a finite set G is closed under an associative product and both cancellation laws hold, then G is a group.

Proof. Let G be a finite set that is closed under an associative product and satisfies both cancellation laws:

Existence of Identity: There exists an element $e \in G$ such that for all $a \in G$, $a \cdot e = e \cdot a = a$.

Existence of Inverse: For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$. Therefore, G with the defined product operation, associative property, closure, identity element e , and inverses for all elements forms a group.

Hence, if a finite set G is closed under an associative product and both cancellation laws hold, then G is a group.

4. Let $G = \{1, -1, i, -i\}$ where i is the imaginary unit with $i^2 = -1$. Define the operation as multiplication of complex numbers. Determine whether G forms a group under this operation. If it does, verify whether G is abelian.

Closure: To verify closure under multiplication of complex numbers:

$$\begin{aligned} 1 \cdot 1 &= 1, \\ 1 \cdot (-1) &= -1, \\ 1 \cdot i &= i, \\ 1 \cdot (-i) &= -i, \\ (-1) \cdot (-1) &= 1, \\ (-1) \cdot i &= -i, \\ (-1) \cdot (-i) &= i, \\ i \cdot i &= -1, \\ i \cdot (-i) &= -i, \\ (-i) \cdot (-i) &= -1. \end{aligned}$$

Since each product $g \cdot h$ is in G , G is closed under the defined operation.

Identity Element: The element 1 serves as the identity:

$$1 \cdot g = g \cdot 1 = g \quad \text{for all } g \in G.$$

Inverses:

$$1^{-1} = 1, \quad (-1)^{-1} = -1, \quad i^{-1} = -i, \quad (-i)^{-1} = i.$$

Each element in G has an inverse within G .

Associativity: Multiplication of complex numbers is associative.

Abelianness:

$$1 \cdot g = g \cdot 1 \quad \text{for all } g \in G.$$

Since all products commute, G is abelian.

Therefore, $G = \{1, -1, i, -i\}$ forms a group under multiplication of complex numbers and is abelian.

Chapter 2

Subgroups

Before delving into the study of groups, we will slightly modify our notation for simplicity. Using the $\cdot$ for the group operation can be cumbersome, so we will henceforth omit it. Instead of writing $a \cdot b$ for elements a and b in G , we will simply write ab .

In general, we will not focus on arbitrary subsets of a group G , as they do not necessarily reflect the algebraic structure of G . Instead, we will consider subsets that possess algebraic properties derived from those of G .

2.1 Definition of a Subgroup

Definition: A nonempty subset H of a group G is a subgroup, denoted as $H \leq G$, if H itself forms a group under the operation defined on G .

Subgroups are important because they retain the group operation from G and help us understand the structure and properties of the larger group. They are key to many important theorems and concepts in group theory.

Remark: If H is a subgroup of G and K is a subgroup of H , then K is a subgroup of G .

Lemma 1: A nonempty subset H of the group G is a subgroup of G if and only if:

1. $a, b \in H$ implies that $ab \in H$.
2. $a \in H$ implies that $a^{-1} \in H$.

Proof.

Given that $H \leq G$, for any $a, b \in H$, the product ab is also in H , satisfying (1). Moreover, each element $a \in H$ has an inverse a^{-1} in H , satisfying (2). Therefore, H fulfills both conditions required to be a subgroup of G .

Let H be a nonempty subset of G such that for all $a, b \in H$, $ab \in H$ and for all $a \in H$, $a^{-1} \in H$.

Claim: $H \leq G$

Existence of Identity: Since H is nonempty, there exists $e \in H$ such that $ea = ae = a$ for all $a \in H$

Closure under Inverses: For each $a \in H$, $a^{-1} \in H$ ensures closure under inverses.

Closure under Group Operation: For $a, b \in H$, $ab \in H$ follows from closure under the group operation of G .

Thus, H satisfies all the criteria to be a subgroup of G . Therefore, a nonempty subset H of the group G is a subgroup of G if and only if conditions (1) and (2) hold.

Lemma 2: If $H \subseteq G$ is a nonempty finite subset of a group G and H is closed under the group

operation of G , then H is a subgroup of G .

Proof.

Since H is closed under the group operation and nonempty, we need to show that H satisfies the subgroup criteria:

Existence of Identity: Since H is nonempty and closed under the group operation, there exists an identity element $e \in H$ such that $eh = he = h$ for all $h \in H$.

Closure under Inverses: For each $h \in H$, since H is closed under the group operation, $h^{-1} \in H$.

Closure under Group Operation: For $h_1, h_2 \in H$, $h_1 h_2 \in H$ follows from the closure of H under the group operation of G .

Thus, H satisfies all the criteria to be a subgroup of G .

Therefore, if $H \subseteq G$ is a nonempty finite subset of a group G and H is closed under the group operation of G , then H is a subgroup of G .

2.2 Some Examples of Subgroups

1. Subgroup of Non-Zero Rational Numbers under Multiplication

$\mathbb{Q}^*$, the set of all non-zero rational numbers.

$\langle \frac{1}{2} \rangle = \{ (\frac{1}{2})^n \mid n \in \mathbb{Z} \}$, the set of all powers of $\frac{1}{2}$.

Closure: For any $(\frac{1}{2})^m, (\frac{1}{2})^n \in \langle \frac{1}{2} \rangle$, $(\frac{1}{2})^m \cdot (\frac{1}{2})^n = (\frac{1}{2})^{m+n} \in \langle \frac{1}{2} \rangle$.

Associativity: Multiplication of rational numbers is associative.

Identity Element: 1 is the identity element in $\mathbb{Q}^*$, and $1 \in \langle \frac{1}{2} \rangle$.

Inverse Element: For any $(\frac{1}{2})^n \in \langle \frac{1}{2} \rangle$, $(\frac{1}{2})^{-n} = 2^n \in \langle \frac{1}{2} \rangle$.

Therefore, $\langle \frac{1}{2} \rangle$ is a subgroup of $(\mathbb{Q}^*, \cdot)$.

2. Subgroup of Symmetric Group

$S_3 = \{1, 2, 3\}$

$H = \{e, (12)\}$

Closure: $(12)(12) = e$ (identity permutation) and $e(12) = (12)e = (12)$.

Associativity: Composition of permutations is associative.

Identity Element: e is the identity permutation.

Inverse Element: $(12)^2 = e$.

Therefore, $\{e, (12)\}$ is a subgroup of S_3 .

2.3 Cyclic Groups

Cyclic groups are fundamental in group theory, characterized by their generation through a single element.

Definition: A cyclic group G is generated by an element a , where every element can be expressed as a^n , for n ranging over all integers.

Example: $\mathbb{Z}_{10}$ is cyclic. In $\mathbb{Z}_{10}$: One example of a generator (or a cyclic element) is 1. Let's verify:

$$1^1 = 1$$

$$1^2 = 1$$

$$1^3 = 1$$

...

$$1^{10} = 1 \text{ (since we are in } \mathbb{Z}_{10})$$

Therefore, 1 generates all elements in $\mathbb{Z}_{10}$. Hence, $\mathbb{Z}_{10}$ is cyclic, and $\langle 1 \rangle = \mathbb{Z}_{10}$.

2.4 Equivalence Relations

Equivalence Relation:

An equivalence relation $\sim$ on a set G satisfies the following properties for elements $a, b, c \in G$:

- Reflexivity: $a \sim a$
- Symmetry: If $a \sim b$, then $b \sim a$
- Transitivity: If $a \sim b$ and $b \sim c$, then $a \sim c$

Equivalence Class: Let G be a group and $\sim$ an equivalence relation on G . The equivalence class of an element $a \in G$ is denoted by $[a]$ and defined as:

$$[a] = \{g \in G \mid g \sim a\}$$

This means $[a]$ consists of all elements in G that are equivalent to a under the relation $\sim$.

Example: Consider the group $\mathbb{Z}$ and the equivalence relation $\equiv \pmod{n}$ (congruence modulo n). The equivalence class of an integer a is:

$$[a] = \{b \in \mathbb{Z} \mid a \equiv b \pmod{n}\}$$

Lemma 3: Equivalence Classes are either disjoint or equal.

Proof. Let G be a set and $\sim$ an equivalence relation on G .

Disjoint Equivalence Classes:

Let $x, y \in G$ are such that $[x] \neq [y]$. To prove that $[x] \cap [y] = \emptyset$, consider any element $z \in [x] \cap [y]$. By definition of intersection, $z \in [x]$ and $z \in [y]$.

Since $[x]$ and $[y]$ are equivalence classes, $z \sim x$ and $z \sim y$. By the properties of equivalence relations:

- $z \sim x$ implies $[z] = [x]$.
- $z \sim y$ implies $[z] = [y]$.

Therefore, $[x] = [y]$, contradicting our assumption $[x] \neq [y]$. Hence, $[x] \cap [y] = \emptyset$, showing that disjoint equivalence classes cannot share any elements. *Equal Equivalence Classes:*

Now suppose $[x] = [y]$. To prove equality, take any $z \in [x]$. Since $[x] = [y]$, $z \sim x$, which implies $z \sim y$. Hence, $z \in [y]$.

Conversely, take any $w \in [y]$. Since $[y] = [x]$, $w \sim y$, which implies $w \sim x$. Hence, $w \in [x]$.

Therefore, $[x] \subseteq [y]$ and $[y] \subseteq [x]$, proving $[x] = [y]$.

Equivalence Classes under an equivalence relation $\sim$ on a set G are either disjoint or equal. This property is fundamental in the partitioning of G into distinct subsets based on their equivalence under $\sim$.

Lemma 4: Equivalence Classes partition the set.

Proof. Let $S = \{[a] \mid a \in A\}$, where A is a nonempty set and $\sim$ is an equivalence relation on A .

$$[a] = \{x \in A \mid x \sim a\} \text{ for a nonempty set } A, a \in A.$$

Since $\sim$ is an equivalence relation:

For all $a \in A$, $a \sim a$ (reflexive property), hence $a \sim [a]$, so $[a] \neq \emptyset$.

Take $x \in A$. Then $x \sim [x]$ (reflexive property), meaning x belongs to at least one equivalence class.

Let $x \in [a]$ and $x \in [b]$. Then $x \sim a$ and $x \sim b$. $a \sim x$ and $x \sim b$ imply $a \sim b$, which implies $[a] = [b]$.

This shows that x must belong to a unique equivalence class. Thus, S partitions the set A .

Definition: Let G be a group, H a subgroup of G . For $a, b \in G$, we say a is congruent to b mod H , written as $a \equiv b \pmod{H}$, if $ab^{-1} \in H$.

Lemma 5: The relation $a \equiv b \pmod{H}$ is an equivalence relation.

Proof. To prove that $\equiv \pmod{H}$ is an equivalence relation, we need to verify three properties:

Reflexivity: For any $a \in G$, $a \equiv a \pmod{H}$. This holds because $a^{-1}a = e \in H$, so $a \equiv a \pmod{H}$.

Symmetry: If $a \equiv b \pmod{H}$, then $b \equiv a \pmod{H}$. This follows since $ab^{-1} \in H$ implies $(ab^{-1})^{-1} = (b^{-1}a)^{-1} = a^{-1}(b^{-1})^{-1} = a^{-1}b \in H$, hence $b \equiv a \pmod{H}$.

Transitivity: If $a \equiv b \pmod{H}$ and $b \equiv c \pmod{H}$, then $a \equiv c \pmod{H}$. Suppose $ab^{-1} \in H$ and $bc^{-1} \in H$. Then $(ab^{-1})(bc^{-1}) = a(b^{-1}b)c^{-1} = aec^{-1} = ac^{-1} \in H$, so $a \equiv c \pmod{H}$.

Therefore, $\equiv \pmod{H}$ satisfies the reflexive, symmetric, and transitive properties, confirming it is an equivalence relation.

2.5 Cosets

Definition: Let G be a group and $H \leq G$ be a subgroup. Then, for $a \in G$:

$Ha = \{ha \mid h \in H\}$ is called the right coset of H in G

$aH = \{ah \mid h \in H\}$ is called the left coset of H in G

Remark: Equivalence classes and cosets both use partitioning principles to define quotient structures. Equivalence classes group elements by shared properties, while cosets partition groups by subgroup operations, highlighting symmetry in group theory. Together, they are powerful tools for understanding relationships and symmetry in mathematics.

Properties of Right Cosets

Let H be a subgroup of G , and let $a, b \in G$. Then:

1. $a \in Ha$,
2. $Ha = H$ if and only if $a \in H$,
3. $Ha = Hb$ if and only if $a \in Hb$,
4. $Ha \subseteq Hb$ if and only if $ab^{-1} \in H$,
5. $Ha = Hb$ if and only if $a^{-1}b \in H$,
6. $|Ha| = |Hb|$,
7. $Ha = aH$ if and only if $H = aHa^{-1}$,
8. Ha is a subgroup of G if and only if $a \in H$.

Proof.

1. $a = ea \in Ha$.
2. To prove property 2, suppose $Ha = H$. Then $a \in Ha = H$. Conversely, if $a \in H$, then $Ha \subseteq H$ and $H \subseteq Ha$ follows from H 's closure.
3. If $Ha = Hb$, then $a \in Ha = Hb$. Conversely, if $a \in Hb$, then $a = bh$ for some $h \in H$, so $Ha = H(bh^{-1}) = Hb$.
4. Property 4 follows from property 3, since if $c \in Ha \cap Hb$, then $cH = Ha$ and $cH = Hb$.
5. $Ha = Hb$ if and only if $a^{-1}b \in H$, which follows from property 2.
6. To show $|Ha| = |Hb|$, define a one-to-one mapping $ha \mapsto hb$ from Ha to Hb , which is one-to-one by the cancellation property.
7. $Ha = aH$ if and only if $(Ha)a^{-1} = (aH)a^{-1} = H$, hence $aHa^{-1} = H$.
8. If Ha is a subgroup, it contains the identity e , so $Ha \supseteq He = H$. By property 4, $Ha = eH = H$. Thus, if $a \in H$, then $Ha = H$, and conversely, if $Ha = H$, then $a \in H$.

Lemma 6: For all $a \in G$,

$$Ha = \{x \in G \mid a \equiv x \pmod{H}\}.$$

Proof. Let $[a] = \{x \in G \mid a \equiv x \pmod{H}\}$. We first show that $Ha \subseteq [a]$.

For $h \in H$, we have

$$a(ha)^{-1} = a(a^{-1}h^{-1}) = h^{-1} \in H$$

since H is a subgroup of G . By the definition of congruence mod H , this implies that $ha \in [a]$ for every $h \in H$. Thus, $Ha \subseteq [a]$.

Now, suppose $x \in [a]$. This means $ax^{-1} \in H$, so

$$(ax^{-1})^{-1} = xa^{-1}$$

is also in H . Therefore, $xa^{-1} = h$ for some $h \in H$. Multiplying both sides by a , we get

$$x = ha,$$

and thus $x \in Ha$. Therefore, $[a] \subseteq Ha$.

Since we have shown both $[a] \subseteq Ha$ and $Ha \subseteq [a]$, we conclude that

$$[a] = Ha,$$

which is the assertion of the lemma.

Lemma 7: There is a one-to-one correspondence between any two right cosets of a subgroup H in G .

Proof. Let aH and bH be arbitrary right cosets of H in G . Define a mapping $\phi : aH \rightarrow bH$ by $\phi(x) = bx$ for all $x \in aH$.

Well-defined: Suppose $x \in aH$. Then $x = ah$ for some $h \in H$. Hence, $\phi(x) = bx = b(ah) = (ba)h \in bH$, showing $\phi(x) \in bH$ does not depend on the choice of representative ah .

Injectivity: Suppose $\phi(x_1) = \phi(x_2)$. Then $bx_1 = bx_2$, implying $b(x_1x_2^{-1}) = e$, where e is the identity in G . Therefore, $x_1 = x_2$, proving ϕ is injective.

Surjectivity: For any $y \in bH$, there exists $x \in aH$ such that $\phi(x) = y$ (specifically, $x = b^{-1}y \in aH$), proving ϕ is surjective.

Thus, ϕ is a well-defined bijection between aH and bH , establishing a one-to-one correspondence between any two right cosets of H in G .

2.6 Some Examples of Cosets

1. Cosets in $\mathbb{Z}/6\mathbb{Z}$

Group: $(\mathbb{Z}/6\mathbb{Z}, +_6)$, where $\mathbb{Z}/6\mathbb{Z}$ denotes integers modulo 6.

Subgroup: $H = \{0, 2, 4\} +_6$, the subgroup of elements 0, 2, 4 modulo 6.

$$0 +_6 H = \{0 +_6 0, 0 +_6 2, 0 +_6 4\} = \{0, 2, 4\}$$

$$1 +_6 H = \{1 +_6 0, 1 +_6 2, 1 +_6 4\} = \{1, 3, 5\}$$

$$2 +_6 H = \{2 +_6 0, 2 +_6 2, 2 +_6 4\} = \{2, 4, 0\} = \{0, 2, 4\}$$

2. Cosets in S_4

Group: S_4 , the symmetric group on $\{1, 2, 3, 4\}$.

Subgroup: $H = \langle (1234) \rangle$, where (1234) is a cycle permutation.

$$eH = \{e \cdot e, e \cdot (1234)\} = \{e, (1234)\}$$

$$(12)H = \{(12) \cdot e, (12) \cdot (1234)\} = \{(12), (12)(1234)\}$$

$$(34)H = \{(34) \cdot e, (34) \cdot (1234)\} = \{(34), (34)(1234)\}$$

2.7 Lagrange's Theorem

Index of a Group: If H is a subgroup of G , the index of H in G , denoted $[G : H]$ or $i_G(H)$, is the number of distinct right cosets of H in G .

If G is a finite group, then for H a subgroup of G :

$$i_G(H) = \frac{|G|}{|H|}.$$

THEOREM 1: Let G be a finite group and H be a subgroup of G . Then the order of H , denoted $|H|$, divides the order of G , denoted $|G|$.

$$|G| = [G : H] \cdot |H|,$$

where $[G : H]$ denotes the index of H in G , the number of distinct cosets of H in G .

Proof. Let $aH = \{ha \mid h \in H\}$ denote the right coset of H generated by a . These cosets form an equivalence relation $\sim$ on G , where each equivalence class Ha is disjoint.

Since the cosets Ha partition G , we have:

$$G = \bigcup_{i=1}^n Ha_i$$

where Ha_i are distinct right cosets of H .

Each coset Ha_i has the same cardinality as H , denoted $|H|$. Therefore, the order of G , denoted $|G|$, is the sum of the cardinalities of these cosets:

$$|G| = |Ha_1| + |Ha_2| + \dots + |Ha_n| = n \cdot |H|.$$

Thus, $|G|$ is a multiple of $|H|$, which means $|H|$ divides $|G|$:

$$|G| = k \cdot |H|$$

for some integer k .

Therefore, $|H|$ divides $|G|$, as required.

Lagrange's theorem has some very important corollaries. Before we present these we make one definition.

Definition: If G is a group and $a \in G$, the *order (or period)* of a is the least positive integer m such that $a^m = e$, where e is the identity element of G .

Corollary 1: If G is a finite group and $a \in G$, then $o(a) \mid |G|$.

Proof. Let $\langle a \rangle$ denote the cyclic subgroup of G generated by a , consisting of elements $e, a, a^2, \dots, a^{o(a)-1}$, where $o(a)$ is the order of a . Since $a^{o(a)} = e$, $\langle a \rangle$ contains $o(a)$ elements.

By Lagrange's theorem, the order of any subgroup divides the order of the group. Therefore, $o(a) \mid |G|$, where $|G|$ represents the order of the group G . Thus, $o(a)$, the order of a , divides the order of G , as required.

Corollary 2: If G is a finite group and $a \in G$, then $a^{|G|} = e$.

Proof. By Corollary 1, $|G| = m \cdot o(a)$ for some integer m . Therefore,

$$a^{|G|} = a^{m \cdot o(a)} = (a^{o(a)})^m = e^m = e.$$

Corollary 3 (Euler's Theorem): If n is a positive integer and a is coprime to n , then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Proof. Euler's theorem is a direct consequence of the properties of Euler's totient function $\phi(n)$, which counts integers coprime to n . By Lagrange's theorem, we know that the order of any subgroup divides the order of the group.

Earlier, we established that for a finite group G and a subgroup H , the index of H in G , denoted as $[G : H]$, represents the number of distinct right cosets of H in G . Furthermore, Corollary 1 and Corollary 2 demonstrated that for any element a in a finite group G , $o(a)$ divides $o(G)$. Now, applying these concepts to number theory, Euler's theorem extends the understanding to modular arithmetic. Specifically, for any integer a coprime to n ,

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

Corollary 4 (Fermat's Little Theorem): If p is a prime number and a is any integer, then $a^p \equiv a \pmod{p}$.

Proof. To apply Corollary 2, consider a modulo p . If a is relatively prime to p , then by Corollary 3 (Euler's theorem), $a^{\phi(p)} \equiv 1 \pmod{p}$. Since $\phi(p) = p - 1$, we have $a^{p-1} \equiv 1 \pmod{p}$. Multiplying both sides by a , we get $a^p \equiv a \pmod{p}$.

If a is not relatively prime to p , then p divides a . Hence, $a \equiv 0 \pmod{p}$, and consequently $a^p \equiv 0 \equiv a \pmod{p}$.

Thus, $a^p \equiv a \pmod{p}$ holds true in all cases, proving Fermat's Little Theorem.

Corollary 5: If G is a finite group whose order is a prime number p , then G is a *cyclic group*.

Proof. Let G be a finite group with $|G| = p$, where p is a prime number. We will show that G is cyclic.

Let H is a subgroup of G . By Lagrange's Theorem, $|H|$ must divide $|G| = p$. The only divisors of p are 1 and p . Therefore, $|H| = 1$ or $|H| = p$.

If $|H| = 1$, then $H = \{e\}$, the trivial subgroup.

If $|H| = p$, then $H = G$ since the order of a subgroup equal to the order of the group implies the subgroup is the entire group.

Consider an element $a \in G$ with $a \neq e$. Let $H = \langle a \rangle$ be the cyclic subgroup generated by a . Since H is a subgroup and $a \neq e$, H is non-trivial, so $|H| = p$. Hence, $H = G$.

Thus, every element of G can be expressed as a power of a , showing that G is cyclic with a as a generator.

Therefore, if $|G| = p$ and p is a prime, G is a cyclic group.

2.8 A Counting Principle

Lemma 8: HK is a subgroup of G if and only if $HK = KH$.

Proof. Let $HK = KH$

Claim: HK is a subgroup

For closure, let $x = hk \in HK$ and $y = h'k' \in HK$. Then $xy = hk \cdot h'k'$. Since $KH = HK$, we can write $kh' = h_2k_2$ with $h_2 \in H$ and $k_2 \in K$. Therefore,

$$xy = h(h_2k_2)k' = (hh_2)(k_2k') \in HK,$$

proving closure.

For inverses, let $x = hk \in HK$. Then

$$x^{-1} = (hk)^{-1} = k^{-1}h^{-1}.$$

Since $KH = HK$, $k^{-1}h^{-1} \in KH = HK$. Hence, $x^{-1} \in HK$. Therefore, HK contains inverses. Thus, HK is a subgroup.

Let HK is a subgroup of G

Let HK is a subgroup. To show $HK = KH$, first note that for any $h \in H$ and $k \in K$, $kh \in HK$. Since HK is a subgroup, $k^{-1}h^{-1} \in HK$. Therefore,

$$(kh)^{-1} = h^{-1}k^{-1} \in HK,$$

implying $kh \in HK$. Thus, $KH \subseteq HK$.

Now, take any $x = hk \in HK$. Since HK is a subgroup,

$$x^{-1} = (hk)^{-1} = k^{-1}h^{-1} \in HK.$$

Then

$$(k^{-1}h^{-1})^{-1} = hk \in KH,$$

implying $x = hk \in KH$. Thus, $HK \subseteq KH$.

Since $KH \subseteq HK$ and $HK \subseteq KH$, we conclude $HK = KH$.

Therefore, HK is a subgroup of G if and only if $HK = KH$.

Corollary 5: If H and K are subgroups of the abelian group G , then HK is a subgroup of G .

Proof. Since G is abelian, the operation is commutative, meaning $gh = hg$ for all $g, h \in G$. This implies that for any $h \in H$ and $k \in K$, $hk = kh$.

We need to show that $HK = \{hk \mid h \in H, k \in K\}$ is a subgroup of G . To do this, we must verify that HK is closed under the group operation and that every element in HK has an inverse in HK .

Closure: Let $x = hk \in HK$ and $y = h'k' \in HK$. Then,

$$xy = hk \cdot h'k' = hkh'k'.$$

Since G is abelian, we can rearrange to get:

$$xy = hh'kk'.$$

Since $hh' \in H$ and $kk' \in K$ (because H and K are subgroups and thus closed under the group operation), we have $hh' \in H$ and $kk' \in K$, so:

$$xy \in HK.$$

Thus, HK is closed under the group operation.

Inverses: Let $x = hk \in HK$. Then,

$$x^{-1} = (hk)^{-1} = k^{-1}h^{-1}.$$

Since G is abelian, we have:

$$k^{-1}h^{-1} = h^{-1}k^{-1}.$$

Since H and K are subgroups, $h^{-1} \in H$ and $k^{-1} \in K$. Therefore,

$$x^{-1} \in HK.$$

Thus, HK contains the inverses of its elements.

Since HK is closed under the group operation and contains the inverses of its elements, HK is a subgroup of G .

THEOREM 2: If H and K are finite subgroups of G of orders $o(H)$ and $o(K)$, respectively, then

$$o(HK) = \frac{o(H)o(K)}{o(H \cap K)}$$

Proof. To prove the theorem, consider the subgroup HK generated by all products hk , where $h \in H$ and $k \in K$.

HK is a subgroup of G because it is closed under the group operation: for any $h, h' \in H$ and $k, k' \in K$, $(hk)(h'k') = (hh')(k(k')^{-1}) \in HK$.

Each element hk in HK is distinct because if $hk = h'k'$, then $h'^{-1}h = k'k^{-1} \in H \cap K$. Therefore, $h = h'$ and $k = k'$, ensuring uniqueness.

The order $o(HK)$ is the number of distinct elements in HK . Since $H \times K$ has $o(H) \cdot o(K)$ elements, and each corresponds uniquely to an element in HK , we have $o(HK) = o(H) \cdot o(K)$.

Consider $H \cap K$, the intersection of H and K . For every $h \in H \cap K$, $hk = h(h^{-1}k)$. This shows that each element hk in HK can be represented uniquely by $h \in H \cap K$. Hence, the number of distinct elements in HK is reduced by $o(H \cap K)$.

Therefore, $o(HK)$, the order of HK , is given by $o(HK) = \frac{o(H) \cdot o(K)}{o(H \cap K)}$.

Corollary 6: If H and K are subgroups of G such that $o(H) > \sqrt{o(G)}$ and $o(K) > \sqrt{o(G)}$, then $H \cap K \neq \{e\}$.

Proof.

Given that $o(H) > \sqrt{o(G)}$ and $o(K) > \sqrt{o(G)}$, we apply Theorem 2.5.1.

From Theorem above, we know:

$$o(HK) = \frac{o(H)o(K)}{o(H \cap K)}$$

Since $o(H) > \sqrt{o(G)}$ and $o(K) > \sqrt{o(G)}$, it follows that:

$$o(H) \cdot o(K) > \sqrt{o(G)} \cdot \sqrt{o(G)} = o(G)$$

Therefore, the order of the subgroup HK is greater than the order of G .

However, HK is a subgroup of G , so its order cannot exceed $o(G)$.

This contradiction implies that our initial assumption must be incorrect. Therefore, we conclude that $H \cap K \neq \{e\}$.

Application: Consider G as a finite group of order pq , where p and q are primes with $p > q$. Applying the corollary, there can be at most one subgroup of order p . If H and K are such subgroups, $H \cap K \neq \{e\}$ implies $H = K$, establishing that there is exactly one subgroup of order p in G . This insight helps in determining the complete structure of G .

2.9 Few Examples

1. Let G be a group, W a subset of G . Let $\langle W \rangle$ be the set of all elements of G representable as a product of elements of W raised to positive, zero, or negative integer exponents. $\langle W \rangle$ is the subgroup of G generated by W and is the smallest subgroup of G containing W . In fact, $\langle W \rangle$ is the intersection of all the subgroups of G which contain W (this intersection is not vacuous since G is a subgroup of G which contains W).

Subgroup Generated by W : Denoted $\langle W \rangle$, it consists of all elements of G that can be expressed as finite products (including inverses) of elements from W . In mathematical terms:

$$\langle W \rangle = \{w_1^{e_1} w_2^{e_2} \cdots w_n^{e_n} \mid w_i \in W, e_i \in \mathbb{Z}, n \in \mathbb{N}\}$$

Here, $\mathbb{Z}$ denotes the set of all integers, $\mathbb{N}$ denotes the set of all natural numbers, and $w_i^{e_i}$ represents the element w_i raised to the power e_i .

Smallest Subgroup Containing W : $\langle W \rangle$ is indeed the smallest subgroup of G that contains W . This means any subgroup of G containing W must also contain $\langle W \rangle$.

Intersection of Subgroups: $\langle W \rangle$ can also be defined as the intersection of all subgroups of G that contain W . This intersection is non-empty because G itself is a subgroup containing W .

2. (a) If H is a subgroup of G , and $a \in G$, let $aHa^{-1} = \{aha^{-1} \mid h \in H\}$. Show that aHa^{-1} is a subgroup of G .

(b) If H is finite, what is $|aHa^{-1}|$? Let H be a subgroup of G , and let $a \in G$. Define $aHa^{-1} = \{aha^{-1} \mid h \in H\}$.

Closure: For any $h_1, h_2 \in H$,

$$ah_1a^{-1} \cdot ah_2a^{-1} = ah_1h_2a^{-1}$$

Since H is closed under its operation, $h_1h_2 \in H$, so $ah_1h_2a^{-1} \in aHa^{-1}$.

Identity element: $aea^{-1} = aa^{-1} = e$, where e is the identity element of G . Thus, $e \in aHa^{-1}$.

Inverse: For any $h \in H$,

$$(aha^{-1})^{-1} = (a^{-1})^{-1}h^{-1}a^{-1} = ah^{-1}a^{-1}$$

Since $h^{-1} \in H$, $ah^{-1}a^{-1} \in aHa^{-1}$.

Therefore, aHa^{-1} satisfies the subgroup criteria and is a subgroup of G .

If H is finite, $|aHa^{-1}|$ denotes the cardinality of the set aHa^{-1} .

Each element $aha^{-1} \in aHa^{-1}$ corresponds uniquely to $h \in H$. Therefore, $|aHa^{-1}| = |H|$.

Thus, $|aHa^{-1}| = |H|$, where $|H|$ is the order (number of elements) of the subgroup H .

3. Let a and b be non-identity elements of different orders in a group G of order 155. Prove that the only subgroup of G that contains a and b is G itself.

Let H be a non-trivial subgroup of G containing both a and b .

By Lagrange's theorem, $|H|$ must divide the order of G . Since $|G| = 155$, the possible orders of H are 1, 5, 31, or 155.

If $|H| = 5$, then H is cyclic and all non-identity elements of H have order 5.

Similarly, if $|H| = 31$, then all non-identity elements of H have order 31.

Since a and b have different orders, neither $|H| = 5$ nor $|H| = 31$ can be true.

Therefore, $|H| = 155$ and $H = G$.

4. Show that U_8 is not a cyclic group.

The group of units modulo 8, denoted U_8 , consists of the elements that are coprime to 8. These elements are:

$$U_8 = \{1, 3, 5, 7\}$$

To determine if U_8 is a cyclic group, we need to see if there exists an element in U_8 that generates the entire group.

The order of 1:

$$1^1 \equiv 1 \pmod{8}$$

So, the order of 1 is 1.

The order of 3:

$$3^1 \equiv 3 \pmod{8}$$

$$3^2 \equiv 9 \equiv 1 \pmod{8}$$

So, the order of 3 is 2.

The order of 5:

$$5^1 \equiv 5 \pmod{8}$$

$$5^2 \equiv 25 \equiv 1 \pmod{8}$$

So, the order of 5 is 2.

The order of 7:

$$7^1 \equiv 7 \pmod{8}$$

$$7^2 \equiv 49 \equiv 1 \pmod{8}$$

So, the order of 7 is 2.

Since none of the elements in U_8 have an order of 4, which is the size of the group, there is no element in U_8 that can generate the entire group. Therefore, U_8 is not a cyclic group.

Chapter 3

Normal Subgroups and Quotient Groups

In group theory, not all subgroups H of a group G have the property that left cosets aH equal right cosets Ha for every $a \in G$. However, recognizing those subgroups where this equality holds has been of critical importance since Galois identified them as worthy of special attention approximately 175 years ago.

3.1 Normal Subgroups

Definition: A subgroup N of a group G is said to be a *normal subgroup* of G if for every $g \in G$ and $n \in N$, the element $gng^{-1} \in N$.

Lemma 1: N is a normal subgroup of G if and only if $gNg^{-1} = N$ for every $g \in G$.

Proof.

N is a normal subgroup of $G \Rightarrow gNg^{-1} = N$ for every $g \in G$

Let N is a normal subgroup of G . By definition, for every $g \in G$, $gNg^{-1} = N$.

$gNg^{-1} = N$ for every $g \in G \Rightarrow N$ is a normal subgroup of G

Assume $gNg^{-1} = N$ for every $g \in G$.

Now, to prove that N is normal, we need to show $gNg^{-1} \subseteq N$ and $gNg^{-1} \supseteq N$ for all $g \in G$.

$gNg^{-1} \subseteq N$: Given $gNg^{-1} = N$, for any $n \in N$, there exists some $n' \in N$ such that $gng^{-1} = n'$.

Hence, $gng^{-1} \in N$.

$gNg^{-1} \supseteq N$: Conversely, for any $n \in N$, we have $gng^{-1} = n'$ for some $n' \in N$, which implies $gng^{-1} \in N$.

Therefore, $gNg^{-1} = N$ implies that N is closed under conjugation by any element $g \in G$, which is the definition of being a normal subgroup.

Hence, we have shown both directions of the proof, establishing that N is a normal subgroup of G if and only if $gNg^{-1} = N$ for every $g \in G$.

Lemma 2: The subgroup N of G is a normal subgroup of G if and only if every left coset of N in G is a right coset of N in G .

Proof. Let N is a normal subgroup of G .

Let H be a normal subgroup of G . Then, for all $g \in G$ and $h \in H$,

$$ghg^{-1} \in H.$$

Let $x \in ghg^{-1}$. Since $ghg^{-1} \in H$, it follows that $x \in H$.

Now, consider any $y \in H$. We have $g^{-1}yg \in H$ because H is normal in G .

Therefore, $y = g^{-1}(ghg^{-1})g$. Since $h \in H$, $y = g^{-1}hg \in H$.

Thus, $H \subseteq gHg^{-1}$.

Now, consider any $y \in gHg^{-1}$. Then, $y = g(h'g^{-1})$ for some $h' \in H$.

Hence, $y = gh'g^{-1}$ and $y \in H$.

Therefore, $gHg^{-1} \subseteq H$.

Consequently, $gHg^{-1} = H$.

Now, let $gN = Ng$ for every $g \in G$

For any $g \in G$, the left coset gN is also a right coset. Therefore, there exists some $h \in G$ such that $gN = Nh$.

Now, consider gng^{-1} for $n \in N$:

$$gng^{-1} = gng^{-1}g \cdot g^{-1} = (gng^{-1})g$$

Since $gN = Nh$, there exists $n' \in N$ such that $gng^{-1} = nh$.

Therefore, $gng^{-1} \in N$ for all $g \in G$ and $n \in N$, which proves that N is normal in G .

Hence proved.

Lemma 3: A subgroup N of G is a normal subgroup of G if and only if the product of two right cosets of N in G is again a right coset of N in G .

Proof.

Let N be a normal subgroup of G . Then for any $g \in G$ and $n \in N$, $gng^{-1} \in N$.

Let g_1N and g_2N be right cosets of N in G . Their product is:

$$g_1Ng_2N = (g_1g_2)N$$

Since N is normal, $g_1g_2N = (g_1g_2)N$, which is a right coset of N in G .

Now let $g_1N \cdot g_2N = (g_1g_2)N$, is again a right coset of N in G .

For any $g \in G$ and $n \in N$, consider gng^{-1} :

$$gng^{-1} = gng_1g_1^{-1}g_1g_2g_2^{-1} = (gng_1)(g_1^{-1}g_1g_2)g_2^{-1}$$

Since $g_1N \cdot g_2N = (g_1g_2)N$, $g_1g_2N = (g_1g_2)N$, implying $gng^{-1} \in N$.

Therefore, N is normal in G .

Hence proved.

3.2 Quotient Groups

THEOREM 1: If G is a group and N is a normal subgroup of G , then G/N is also a group. It is called the quotient group or factor group of G by N .

Proof. To show that G/N is a group, we need to verify the following:

Closure: For any $gN, hN \in G/N$, the product $(gN)(hN) = (gh)N$ is well-defined and belongs to G/N . Since N is normal in G , $gh \in G$ and thus $(gh)N \in G/N$.

Associativity: The operation $(gN)(hN) = (gh)N$ inherits associativity from G .

Identity Element: The identity element in G/N is $eN = N$, where e is the identity element of G .

For any $gN \in G/N$,

$$(gN)(eN) = (ge)N = gN$$

$$(eN)(gN) = (eg)N = gN$$

Therefore, eN acts as the identity element in G/N .

Inverse Element: For any $gN \in G/N$, the inverse element is $(gN)^{-1} = (g^{-1})N$. This is because

$$(gN)(g^{-1}N) = (gg^{-1})N = eN = N$$

$$(g^{-1}N)(gN) = (g^{-1}g)N = eN = N$$

Thus, G/N satisfies all the properties of a group, confirming that G/N is indeed a group. It is known as the quotient group or factor group of G by N .

Lemma 4: If G is a finite group and N is a normal subgroup of G , then $|G/N| = \frac{|G|}{|N|}$.

Proof. Since N is a normal subgroup of G , the quotient group G/N is well-defined and finite because G is finite.

By definition, $|G/N|$, the order of the quotient group G/N , is equal to the index of N in G , denoted as $[G : N]$, which is the number of distinct cosets of N in G .

According to Lagrange's theorem:

$$|G| = |N| \cdot |G/N|$$

Dividing both sides by $|N|$, we get:

$$|G/N| = \frac{|G|}{|N|}$$

Thus, $|G/N| = \frac{|G|}{|N|}$.

3.3 Few Examples

1. If G is a group and H is a subgroup of index 2 in G , prove that H is a normal subgroup of G .

Let G be a group and H a subgroup of G with index 2. This means that there are exactly two distinct left cosets of H in G , which we can denote as H and gH for some $g \in G$.

Since the index of H in G is 2, the left cosets and the right cosets of H must coincide, meaning G can be partitioned into two right cosets as well: H and Hg .

Case 1: $x \in H$:

$$xHx^{-1} = Hx^{-1} \subseteq H$$

Since H is a subgroup, $x^{-1} \in H$ and thus $Hx^{-1} = H$.

Case 2: $x \in gH$ where $g \notin H$:

$$x = gh \text{ for some } h \in H$$

Then,

$$xHx^{-1} = (gh)H(gh)^{-1} = ghHh^{-1}g^{-1} = gHg^{-1} \subseteq G$$

Since $x = gh \in gH$ is arbitrary, gHg^{-1} must form a subgroup of G . As there are only two cosets and both must have the same order, it must also be true that $gHg^{-1} = H$. Hence, H is normal in G .

Thus, H is a normal subgroup of G .

2. Suppose H is the only subgroup of order $|H|$ in the finite group G . Prove that H is a normal subgroup of G .

Let $g \in G$ be any element. Consider the conjugate subgroup gHg^{-1} . The order of the subgroup gHg^{-1} is the same as the order of H , i.e., $|gHg^{-1}| = |H|$.

Since H is the only subgroup of G with order $|H|$, it follows that gHg^{-1} must be equal to H . In other words,

$$gHg^{-1} = H \quad \text{for all } g \in G.$$

Hence, H is a normal subgroup of G .

3. If a cyclic subgroup T of G is normal in G , then show that every subgroup of T is normal in G .

Let G be a group and T a cyclic subgroup of G that is normal in G . This means $gTg^{-1} = T$ for all $g \in G$.

Let H be any subgroup of T . We need to show that H is normal in G , i.e., $gHg^{-1} = H$ for all $g \in G$.

Since T is cyclic, there exists an element $t \in G$ such that $T = \langle t \rangle$. Let H be generated by t^k for some integer k , i.e., $H = \langle t^k \rangle$.

For any $g \in G$, we have:

$$gHg^{-1} = g\langle t^k \rangle g^{-1} = \langle gt^k g^{-1} \rangle$$

Since T is normal in G , we know $gTg^{-1} = T$. Therefore, $gtg^{-1} \in T$.

Because $gtg^{-1} \in T$ and T is cyclic, $gtg^{-1} = t^m$ for some integer m .

Let $h \in H$. Since H is generated by t^k , we have $h = (t^k)^n = t^{kn}$ for some integer n .

Then,

$$ghg^{-1} = g(t^{kn})g^{-1} = (gtg^{-1})^{kn} = (t^m)^{kn} = t^{mkn}$$

Since $t^{mkn} \in \langle t^k \rangle = H$, it follows that $gHg^{-1} \subseteq H$.

By similar arguments for the inverse, we have $g^{-1}Hg \subseteq H$.

Therefore,

$$gHg^{-1} = H$$

This shows that H is normal in G .

Hence, if a cyclic subgroup T of G is normal in G , then every subgroup of T is normal in G .

4. Let G be a group in which, for some integer $n > 1$, $(ab)^n = a^n b^n$ for all $a, b \in G$. Show that

(a) $G^{(n)} = \{x^n \mid x \in G\}$ is a normal subgroup of G .

(b) $G^{(n!)} = \{x^{n!} \mid x \in G\}$ is a normal subgroup of G .

(a) *Identity*: The identity element $e \in G$ satisfies $e^n = e$, so $e \in G^{(n)}$.

Closure: For any $a, b \in G$, $a^n, b^n \in G^{(n)}$.

$$(a^n b^n) = (ab)^n \quad (\text{given that } (ab)^n = a^n b^n)$$

Therefore, $(ab)^n \in G^{(n)}$.

Inverses: For any $a \in G$, $a^n \in G^{(n)}$. We need to show that $(a^n)^{-1} \in G^{(n)}$:

$$(a^n)^{-1} = (a^{-1})^n$$

Therefore, $(a^n)^{-1} \in G^{(n)}$.

Next, we show that $G^{(n)}$ is normal in G : For any $g \in G$ and $x^n \in G^{(n)}$,

$$gx^n g^{-1} = (g x g^{-1})^n$$

Since $gxg^{-1} \in G$, $(gxg^{-1})^n \in G^{(n)}$.

Therefore, $gx^n g^{-1} \in G^{(n)}$, which means $G^{(n)}$ is normal in G .

(b) *Identity*: The identity element $e \in G$ satisfies $e^{n!} = e$, so $e \in G^{(n!)}$.

Closure: For any $a, b \in G$, $a^{n!}, b^{n!} \in G^{(n!)}$. We need to show that $(a^{n!}b^{n!}) \in G^{(n!)}$:

$$(a^{n!}b^{n!}) = (ab)^{n!} \quad (\text{since } (ab)^n = a^n b^n \text{ implies } (ab)^{n!} = a^{n!} b^{n!})$$

Therefore, $(ab)^{n!} \in G^{(n!)}$.

Inverses: For any $a \in G$, $a^{n!} \in G^{(n!)}$. We need to show that $(a^{n!})^{-1} \in G^{(n!)}$:

$$(a^{n!})^{-1} = (a^{-1})^{n!}$$

Therefore, $(a^{n!})^{-1} \in G^{(n!)}$.

Next, we show that $G^{(n!)}$ is normal in G : For any $g \in G$ and $x^{n!} \in G^{(n!)}$,

$$gx^{n!}g^{-1} = (gxg^{-1})^{n!}$$

Since $gxg^{-1} \in G$, $(gxg^{-1})^{n!} \in G^{(n!)}$.

Therefore, $gx^{n!}g^{-1} \in G^{(n!)}$, which implies $G^{(n!)}$ is normal in G .

Chapter 4

Homomorphisms

Homomorphisms in group theory serve to preserve the algebraic structure between groups, ensuring that the operation on elements in one group translates consistently to another. They facilitate the study of group relationships and properties, revealing insights into symmetry and structure through concepts like kernels and images. Beyond theoretical applications, homomorphisms play a crucial role in practical fields such as cryptography and physics, illustrating their broad relevance and significance in both pure mathematics and applied sciences.

4.1 Definition of Homomorphism

Definition: A mapping ϕ from a group G into a group C is said to be a homomorphism if for all $a, b \in G$,

$$\phi(ab) = \phi(a) \cdot \phi(b).$$

Remark: In a homomorphism ϕ between groups G and C , $\phi(ab)$ computes the product using G 's operation, while $\phi(a) \cdot \phi(b)$ uses C 's operation, illustrating how ϕ maps operations between the groups while preserving their algebraic structures.

4.2 Kernel

The kernel of a homomorphism in group theory is crucial as it identifies elements in the domain group that map to the identity in the codomain. It serves to study the structure of the group, determines the homomorphism's injectivity and surjectivity, and finds applications in cryptography and algorithm design.

Definition: If ϕ is a homomorphism from a group G to a group G' , the kernel of ϕ , denoted as $\ker(\phi)$, is defined by

$$\ker(\phi) = \{x \in G \mid \phi(x) = e'\},$$

where e' is the identity element of G' .

4.3 Some Important Lemmas

Lemma 1: Suppose G is a group, N a normal subgroup of G . Define $\phi : G \rightarrow G/N$ by $\phi(x) = Nx$ for all $x \in G$. Then ϕ is a homomorphism of G onto G/N .

Proof.

ϕ is onto G/N : For any $X \in G/N$, there exists $y \in G$ such that $X = Ny$. Hence, $X = \phi(y)$, proving ϕ is onto.

ϕ preserves multiplication: For $x, y \in G$,

$$\phi(xy) = Nxy = NxNy = \phi(x)\phi(y).$$

Therefore, ϕ preserves the group operation, confirming it is a homomorphism from G onto G/N .

Lemma 2: If ϕ is a homomorphism from G to G' , then:

1. $\phi(e) = e'$, where e and e' are the identity elements of G and G' , respectively.
2. $\phi(x^{-1}) = \phi(x)^{-1}$ for all $x \in G$.

Proof.

1. To prove $\phi(e) = e'$, calculate $\phi(x)e = \phi(x) = \phi(xe) = \phi(x)\phi(e)$. By cancellation in G' , $\phi(e) = e'$.

2. For $\phi(x^{-1}) = \phi(x)^{-1}$, note $e' = \phi(e) = \phi(xx^{-1}) = \phi(x)\phi(x^{-1})$. By the definition of $\phi(x^{-1})$ in G' , we get $\phi(x^{-1}) = \phi(x)^{-1}$.

Therefore, ϕ satisfies both properties (1) and (2), confirming it as a homomorphism.

Lemma 3: If ϕ is a homomorphism from G to G' with kernel K , then K is a normal subgroup of G .

Proof.

Closure: If $x, y \in K$, then $\phi(x) = e$ and $\phi(y) = e$, where e is the identity element of G . Therefore, $\phi(xy) = \phi(x)\phi(y) = ee = e$, so $xy \in K$.

Inverses: If $x \in K$, then $\phi(x) = e$. By Lemma 2, $\phi(x^{-1}) = \phi(x)^{-1} = e^{-1} = e$, so $x^{-1} \in K$. Hence, K is a subgroup of G .

For any $g \in G$ and $k \in K$, $\phi(gkg^{-1}) = \phi(g)\phi(k)\phi(g)^{-1} = \phi(g)e\phi(g)^{-1} = e$. Thus, $gkg^{-1} \in K$, proving K is normal in G .

Therefore, K is a normal subgroup of G .

Lemma 4: If ϕ is a homomorphism of G onto C with kernel K , then the set of all inverse images of $g \in C$ under ϕ in G is given by Kx , where x is any particular inverse image of g in G .

Proof. Let $g \in C$ and $x \in G$ such that $\phi(x) = g$.

For any $k \in K$, $\phi(k) = e$ (identity in C). Consider kx :

$$\phi(kx) = \phi(k)\phi(x) = eg = g.$$

Hence, $kx \in \phi^{-1}(g)$. Therefore, $Kx \subseteq \phi^{-1}(g)$.

Let $y \in \phi^{-1}(g)$, meaning $\phi(y) = g$. Since $\phi(x) = g$, we have:

$$\phi(y) = \phi(x) \implies \phi(yx^{-1}) = \phi(y)\phi(x^{-1}) = gg^{-1} = e.$$

Thus, $yx^{-1} \in K$, which implies $y \in Kx$. Therefore, $\phi^{-1}(g) \subseteq Kx$.

Since $Kx \subseteq \phi^{-1}(g)$ and $\phi^{-1}(g) \subseteq Kx$, we conclude that $\phi^{-1}(g) = Kx$.

4.4 Isomorphism

Isomorphic groups have differently labeled elements, but an isomorphism acts like a dictionary that translates between these labels, enabling analogous computations in both groups. The specific mapping defining the isomorphism is crucial, so it's important to always specify this mapping when

proving groups are isomorphic.

Definition: A homomorphism ϕ from G into C is said to be an isomorphism if ϕ is one-to-one.

Definition: Two groups G and G^* are said to be isomorphic if there is an isomorphism of G onto G^* . In this case, we write $G \cong G^*$.

Proposition 1: $G \cong G$.

The identity mapping $\phi : G \rightarrow G$ defined by $\phi(g) = g$ for all $g \in G$ is an isomorphism.

Proposition 2: If $G \cong G^*$, then $G^* \cong G$.

Let $\phi : G \rightarrow G^*$ be an isomorphism. The inverse mapping $\phi^{-1} : G^* \rightarrow G$ is also an isomorphism since ϕ is bijective.

Proposition 3: If $G \cong G^*$ and $G^* \cong G^{**}$, then $G \cong G^{**}$.

Let $\phi : G \rightarrow G^*$ and $\psi : G^* \rightarrow G^{**}$ be isomorphisms. The composition $\psi \circ \phi : G \rightarrow G^{**}$ is an isomorphism since the composition of bijective mappings is bijective, and the homomorphism property is preserved under composition.

Corollary: A homomorphism ϕ of G into G with kernel K_ϕ is an isomorphism of G into G if and only if $K_\phi = \{e\}$.

Proof.

Let ϕ be an isomorphism. An isomorphism is a bijective homomorphism. Since ϕ is injective (one-to-one), the kernel of ϕ must be $\{e\}$. This is because if $\phi(g) = e$ (where e is the identity element in G), then g must be the identity element in G , i.e., $g = e$. Therefore, $K_\phi = \{e\}$.

Now, let $K_\phi = \{e\}$. To show that ϕ is an isomorphism, we need to prove that ϕ is bijective.

Injective: Let $\phi(g_1) = \phi(g_2)$. Then, $\phi(g_1 g_2^{-1}) = e$. Since the kernel is $\{e\}$, it follows that $g_1 g_2^{-1} = e$, implying $g_1 = g_2$. Therefore, ϕ is injective.

Surjective: Since ϕ is a homomorphism from G onto G , for every $h \in G$, there exists an element $g \in G$ such that $\phi(g) = h$. Therefore, ϕ is surjective.

Since ϕ is both injective and surjective, it is bijective, hence an isomorphism.

4.5 First Isomorphism Theorem

The First Isomorphism Theorem is a fundamental result in group theory that offers deep insights into the structure of groups and their homomorphisms. It establishes a critical relationship between a group, its quotient group, and the image of a homomorphism, enabling a clearer and more simplified understanding of group structures. This theorem is pivotal in many areas of mathematics, as it provides a powerful framework for analyzing and proving the properties of groups by connecting their internal and external symmetries.

THEOREM: Let ϕ be a homomorphism of G onto $\overline{G}$ with kernel K . Then $G/K \cong \overline{G}$.

Proof. Consider the diagram,

$$\begin{array}{ccc} G & \xrightarrow{\phi} & \overline{G} \\ \downarrow \sigma & & \\ G/K & & \end{array}$$

where $\sigma(g) = Kg$.

$$\begin{array}{ccc} G & \xrightarrow{\phi} & \overline{G} \\ \downarrow \sigma & \nearrow \psi & \\ G/K & & \end{array}$$

Define the mapping $\psi : G/K \rightarrow \overline{G}$ by:

if $X \in G/K$, $X = Kg$

then $\psi(X) = \phi(g)$

$$\begin{array}{ccc} g & \longrightarrow & \phi(g) \\ \downarrow & \nearrow \psi & \\ Kg & & \end{array}$$

well defined: If $X \in G/K$,

then $X = Kg = Kkg$ for $k \in K$.

However, if $X = Kg = Kg'$ for $g, g' \in G$,

then $\phi(x) = \langle \phi \rangle(g)$ and also $\phi(x) = \langle \phi \rangle(g')$.

$\psi(X) = \phi(g)$

$\psi(X) = \phi(g')$

let $Kg = Kg'$ then $g = kg'$

$\phi(g) = \phi(kg') = \phi(k)\phi(g') = \phi(g')$

Thus $\phi(g) = \phi(g')$

onto: If $\bar{x} \in \overline{G}$, then $\bar{x} = \phi(g)$ for $g \in G$.

$\bar{x} = \phi(g) = \psi(Kg) = \psi(X)$

Hence, surjective.

onto: Let $\psi(gK_1) = \psi(gK_2)$ for $gK_1, gK_2 \in G/K$.

This implies:

$$\phi(g) = \phi(g),$$

since $\psi(gK_1) = \phi(g)$ and $\psi(gK_2) = \phi(g)$.

Since ϕ is a homomorphism, $\phi(g) = \phi(g)$ holds true.

Now, let's analyze what $gK_1 = gK_2$ implies:

$$gK_1 = gK_2$$

$$g^{-1}gK_1 = g^{-1}gK_2$$

$$K_1 = K_2$$

Therefore, $gK_1 = gK_2$ implies $K_1 = K_2$.

Hence, ψ is injective.

Homomorphism: For $gK, hK \in G/K$,

$$\psi(gK \cdot hK) = \psi((gh)K) = \phi(gh).$$

Since ϕ is a homomorphism,

$$\phi(gh) = \phi(g)\phi(h) = \psi(gK)\psi(hK).$$

Therefore, ψ preserves the group operation, proving ψ is a homomorphism.

Hence, we have established that $G/K \cong G$ under the conditions where ϕ is a homomorphism from G onto G with kernel K .

This establishes a direct link between homomorphic images of a group G and its normal subgroups N : any homomorphic image of G can be expressed as G/N .

Simple groups are then defined as those lacking nontrivial homomorphic images, thereby having no nontrivial normal subgroups.

4.6 Applications

1. Cauchy's Theorem for Abelian Groups

Theorem: Let G be a finite abelian group and p a prime number such that p divides the order of G , denoted $o(G)$. Then, there exists an element $a \in G$ such that $a^p = e$, where e is the identity element of G .

Proof. The theorem is vacuously true for groups with a single element. Let the theorem holds for all abelian groups with fewer elements than G . We aim to prove it for G .

Case 1: G is cyclic of prime order p : G has an element $a \neq e$ such that $a^p = e$.

Case 2: G has a proper subgroup N such that p divides $o(N)$: By induction, N contains an element $b \neq e$ with $b^p = e$. Hence, $b \in G$ satisfies $b^p = e$.

Case 3: p does not divide $o(N)$ for any proper subgroup N of G : G/N is abelian and has an element $XN \neq N$ with $(XN)^p = N$. Therefore, $X^p = e$ in G .

Thus, there exists $a \in G$ such that $a^p = e$.

By induction on the order of G , Cauchy's theorem for abelian groups asserts the existence of an element $a \in G$ such that $a^p = e$ whenever p divides $o(G)$.

2. Sylow's Theorem for Abelian Groups

Theorem: If G is an abelian group of order $o(G)$, and if p is a prime number such that p^k divides $o(G)$ and p^{k+1} does not divide $o(G)$, then G has a subgroup of order p^k .

Proof.

Existence of Sylow p -subgroup: Since G is abelian, all subgroups of G are normal. By the fundamental properties of abelian groups and Sylow's theorems:

There exists at least one subgroup $H \subseteq G$ such that $|H| = p^m$, where $m \geq 0$ and p^m is the highest power of p dividing $o(G)$.

Maximal p -subgroup: Let H be a subgroup of G such that $|H| = p^m$ and m is maximal, meaning p^m is the largest power of p dividing $o(G)$.

Determining $m = k$: Since p^k divides $o(G)$ and p^{k+1} does not divide $o(G)$, $m \leq k$.

Maximality of H : If $m < k$, then there would exist a subgroup of G with order p^{m+1} , contradicting the maximality of m . Therefore, $m = k$.

Hence, H , with order p^k , is a subgroup of G as required by Sylow's theorem for abelian groups.

Therefore, G indeed has a subgroup of order p^k , completing the proof. This subgroup H is unique up to isomorphism and represents the Sylow p -subgroup of G .

4.7 Further Results and Theorems

Corollary: If G is an abelian group of order $o(G)$ and p, q are prime divisors of $o(G)$ such that $p \leq q$, there is a unique subgroup of G of order p^α .

Proof.

Existence of Subgroup: Since G is abelian, by the Fundamental Theorem of Finite Abelian Groups,

G can be expressed as a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \cdots \times \mathbb{Z}_{n_k},$$

where $n_1, n_2, \dots, n_k$ are positive integers such that $n_1 \mid n_2 \mid \cdots \mid n_k$ and $n_1 n_2 \cdots n_k = o(G)$.

Prime Divisors: Let $o(G) = p^\alpha \cdot m$, where p^α is the highest power of p dividing $o(G)$ and $\gcd(p, m) = 1$. Here, p is a prime divisor of $o(G)$.

Structure of G : Considering G is abelian and of order $o(G)$, there exists a subgroup of order p^α , as p^α divides $o(G)$. By the structure theorem, this subgroup can be identified uniquely as:

$$H = \mathbb{Z}_{p^{\alpha_1}} \times \mathbb{Z}_{p^{\alpha_2}} \times \cdots \times \mathbb{Z}_{p^{\alpha_k}},$$

where $\alpha_1 + \alpha_2 + \cdots + \alpha_k = \alpha$ and $\alpha_i \leq \alpha$ for all i .

Uniqueness: To show uniqueness, consider any subgroup $H' \subseteq G$ of order p^α . By Lagrange's theorem, the order of any subgroup of G must divide $o(G) = p^\alpha \cdot m$. Hence, H' must have the form:

$$H' = \mathbb{Z}_{p^{\alpha'_1}} \times \mathbb{Z}_{p^{\alpha'_2}} \times \cdots \times \mathbb{Z}_{p^{\alpha'_k}}$$

with $\alpha'_1 + \alpha'_2 + \cdots + \alpha'_k = \alpha$ and $\alpha'_i \leq \alpha$ for all i .

Since p is a prime and p^{α_i} are the primary cyclic components that make up the subgroup of order p^α , any other subgroup of order p^α in G must have the same cyclic structure, up to isomorphism.

Therefore, H is the unique subgroup of G of order p^α .

Lemma 5: Let $\varphi : G \rightarrow G'$ be a surjective homomorphism with kernel K . For a subgroup $f \subseteq G$, define $H = \{x \in G \mid \varphi(x) \in f\}$.

Proof. H is a subgroup of G :

Closure: Take $x, y \in H$. This implies $\varphi(x), \varphi(y) \in f$. Since f is a subgroup of G , $\varphi(x) \cdot \varphi(y) \in f$. Therefore, $\varphi(xy) = \varphi(x) \cdot \varphi(y) \in f$, implying $xy \in H$. Thus, H is closed under multiplication.

Identity: $\varphi(e) = e'$, where e is the identity in G and e' is the identity in G' . Hence, $e \in H$.

Inverses: If $x \in H$, then $\varphi(x) \in f$. Since f is closed under inverses in G , $\varphi(x)^{-1} \in f$. Thus, $x^{-1} \in H$.

$H \subseteq K$: If $x \in H$, then $\varphi(x) \in f$. By the definition of K as $\ker(\varphi)$, $x \in K$.

If f is normal in G , then H is normal in G : If f is normal in G , for any $g \in G$, $gfg^{-1} \subseteq f$.

Take $h \in H$, so $\varphi(h) \in f$. For $g \in G$,

$$\varphi(ghg^{-1}) = \varphi(g)\varphi(h)\varphi(g^{-1}).$$

Since φ is a homomorphism, $\varphi(ghg^{-1}) = \varphi(g)\varphi(h)\varphi(g)^{-1}$.

$\varphi(h) \in f$ and f is normal, so $\varphi(g)\varphi(h)\varphi(g)^{-1} \in f$.

Hence, $ghg^{-1} \in H$, showing H is normal in G .

One-to-one correspondence:

Define $\psi : \{\text{Subgroups of } G \text{ containing } K\} \rightarrow \{\text{Subgroups of } G'\}$ by $\psi(H) = \varphi(H)$.

ψ is well-defined because if H contains K , then $\varphi(H)$ is a subgroup of G' .

ψ is injective: If $\psi(H_1) = \psi(H_2)$, then $\varphi(H_1) = \varphi(H_2)$. Hence, $H_1 = H_2$ because φ is surjective.

ψ is surjective: For any subgroup $f' \subseteq G'$, $\varphi^{-1}(f')$ is a subgroup of G containing K , and $\psi(\varphi^{-1}(f')) = f'$.

Theorem 2: Let $\varphi : G \rightarrow G'$ be a homomorphism of G onto G' with kernel K , and let N be a normal subgroup of G . Then $G/N \cong (G/K)/(N/K)$.

Proof. Given G and N where $N = \{x \in G \mid \phi(x) \in N'\}$, we first define a homomorphism θ from G

onto G/N by $\theta(g) = Ng$.

Next, we define a mapping $\psi : G \rightarrow G'/N'$ by $\psi(g) = N'\phi(g)$ for all $g \in G$.

To show ψ is onto, take any $g' \in G'$. Since ϕ is onto, there exists some $g \in G$ such that $\phi(g) = g'$.

Hence, any element $N'g'$ in G'/N' can be written as $N'\phi(g) = \psi(g)$, proving ψ is onto.

To verify ψ is a homomorphism, for $a, b \in G$,

$$\psi(ab) = N'\phi(ab)$$

Since ϕ is a homomorphism,

$$\phi(ab) = \phi(a)\phi(b)$$

Therefore,

$$\psi(ab) = N'\phi(a)\phi(b) = N'\phi(a)N'\phi(b) = \psi(a)\psi(b)$$

Hence, ψ is a homomorphism.

Determine the kernel T of ψ : - If $n \in N$, then $\phi(n) \in N'$, so $\psi(n) = N'\phi(n) = N'$, the identity element of G'/N' . Thus, $N \subseteq T$. - Conversely, if $t \in T$, then $\psi(t) = N'$, meaning $N'\phi(t) = N'$.

This implies $\phi(t) \in N'$, hence $t \in N$. Thus, $T \subseteq N$.

Therefore, the kernel T of ψ is N .

Since ψ is a homomorphism of G onto G'/N' with kernel N , by the First Isomorphism Theorem, we have:

$$G/N \cong G'/N'$$

From the First Isomorphism Theorem again, ϕ induces an isomorphism between G/N and $(G/K)/(N'/K)$.

4.8 Commutator of a Group

In group theory, the commutator of two elements a and b in a group G is an element that measures the "difference" between a and b commuting. The commutator of a and b is denoted by $[a, b]$ and is defined as:

$$[a, b] = a^{-1}b^{-1}ab$$

Properties of Commutators:

Identity Element:

$$[a, a] = e$$

Inverse:

$$[a, b] = [b, a]^{-1}$$

Normal Subgroup: The set of all commutators in G forms a normal subgroup of G , known as the commutator subgroup or the derived subgroup, denoted by G' or $[G, G]$.

Abelian Groups: A group G is abelian if and only if all commutators are the identity element, i.e., $[a, b] = e$ for all $a, b \in G$.

Commutator Subgroup

The commutator subgroup G' (also denoted $[G, G]$) is the subgroup generated by all commutators of the group G .

$$G' = \langle [a, b] \mid a, b \in G \rangle$$

Examples of Commutator Group

1. **Symmetric Group S_3 :** In the symmetric group S_3 , the commutator subgroup is A_3 , the alternating group, which consists of the even permutations.
2. **Matrix Groups:** In the general linear group $GL(n, \mathbb{R})$, commutators often play a significant role in studying the structure and properties of the group.

4.9 Few Examples

1. Suppose that p is a prime number greater than 3. Consider the multiplicative group $G = (\mathbb{Z}/p\mathbb{Z})^*$ of order $p - 1$.

- (a) Prove that the set of squares $S = \{x^2 \mid x \in G\}$ is a subgroup of the multiplicative group G .
- (b) Determine the index $[G : S]$.
- (c) Assume that $-1 \notin S$. Then prove that for each $a \in G$, we have either $a \in S$ or $-a \in S$.

(a) $G = (\mathbb{Z}/p\mathbb{Z})^*$

$$S = \{x^2 \mid x \in G\}$$

Non-empty: Since G includes 1, $1^2 = 1 \in S$, so S is non-empty.

Closure: For $a^2, b^2 \in S$ ($a, b \in G$), $(ab)^2 = a^2b^2 \in S$ because $ab \in G$.

Inverses: If $x^2 \in S$, then $(x^2)^{-1} = (x^{-1})^2 \in S$, since $x^{-1} \in G$.

Thus, S is a subgroup of G .

(b) $|G| = p - 1$

and $|S| = \frac{p-1}{2}$, and $[G : S] = \frac{|G|}{|S|} = 2$.

(c) Given $-1 \notin S$, S is a subgroup of index 2 in G . Hence, G is partitioned into cosets S and $G \setminus S$.

For $a \in G$: - If $a \in S$, then $a \in S$. - If $a \notin S$, then $a \in G \setminus S$, and hence $-a = (-1) \cdot a \in S$ because $-1 \in G \setminus S$.

Therefore, for any $a \in G$, either $a \in S$ or $-a \in S$.

2. Let G be the group of positive real numbers under multiplication, and let $(\mathbb{R}, +)$ be the group of all real numbers under addition. Define $\varphi : G \rightarrow \mathbb{R}$ by $\varphi(x) = \log x$.

For $x, y \in G$,

$$\varphi(xy) = \log(xy) = \log(x) + \log(y) = \varphi(x) + \varphi(y).$$

Since the operation on the right side is addition in $\mathbb{R}$, φ is a homomorphism of G into $\mathbb{R}$.

φ is one-to-one and onto, establishing it as an isomorphism between G and $\mathbb{R}$.

3. Let G be any group, and let g be a fixed element in G . Define $\varphi_g : G \rightarrow G$ by $\varphi_g(x) = gxg^{-1}$. Prove that φ_g is an isomorphism of G onto G .

Let G be any group, and let g be a fixed element in G . Define $\varphi_g : G \rightarrow G$ by $\varphi_g(x) = gxg^{-1}$.

$x, y \in G$:

$$\varphi_g(xy) = g(xy)g^{-1} = (gxg^{-1})(gyg^{-1}) = \varphi_g(x)\varphi_g(y).$$

Therefore, φ_g preserves the group operation, hence it is a homomorphism.

Let $\varphi_g(x) = \varphi_g(y)$. Then $gxg^{-1} = gyg^{-1}$, which implies $x = y$. Thus, φ_g is injective.

For any $h \in G$, choose $x = g^{-1}hg$. Then,

$$\varphi_g(x) = g(g^{-1}hg)g^{-1} = h.$$

Hence, φ_g is surjective.

3. (a) Given any group G and a subset U , let H be the smallest subgroup of G which contains U . Prove there is such a subgroup H in G . (This H is called the subgroup generated by U .)
- (b) If $gug^{-1} \in U$ for all $g \in G$ and $u \in U$, prove that H is a normal subgroup of G .
- (a) Let $U \subseteq G$. Consider the set of all subgroups of G that contain U . This set is non-empty since G itself is a subgroup containing U . Let H be the intersection of all these subgroups:

$$H = \bigcap_{\substack{K \leq G \\ U \subseteq K}} K.$$

The intersection of any collection of subgroups is itself a subgroup. Since $U \subseteq K$ for each subgroup K in the intersection, $U \subseteq H$. Therefore, H is a subgroup of G and is the smallest subgroup containing U . Thus, such a subgroup H exists.

- (b) Let $gug^{-1} \in U$ for all $g \in G$ and $u \in U$. Let H be the subgroup generated by U . We need to show that H is normal in G , i.e., $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$.

Since H is the smallest subgroup containing U , every element $h \in H$ can be expressed as a finite product of elements of U and their inverses:

$$h = u_1^{\epsilon_1} u_2^{\epsilon_2} \cdots u_n^{\epsilon_n},$$

where $u_i \in U$ and $\epsilon_i = \pm 1$.

For any $g \in G$,

$$ghg^{-1} = g(u_1^{\epsilon_1} u_2^{\epsilon_2} \cdots u_n^{\epsilon_n})g^{-1}.$$

Using the property $gug^{-1} \in U$,

$$ghg^{-1} = (gu_1^{\epsilon_1}g^{-1})(gu_2^{\epsilon_2}g^{-1}) \cdots (gu_n^{\epsilon_n}g^{-1}).$$

Each term $gu_i^{\epsilon_i}g^{-1} \in U$ or $gu_i^{-\epsilon_i}g^{-1} \in U$ because U is closed under taking inverses and conjugation by g .

Therefore, ghg^{-1} is a product of elements of U , and hence $ghg^{-1} \in H$. Thus, H is normal in G .

4. If N and M are normal subgroups of G , prove that $NM/M \cong N/(N \cap M)$.

Consider the homomorphism $\varphi : N \rightarrow NM/M$ defined by $\varphi(n) = nM$ for $n \in N$.

Since M is a normal subgroup of G , for any $m \in M$ and $n \in N$,

$$\varphi(nm) = (nm)M = n(mm^{-1})M = nM = \varphi(n).$$

For any $n_1, n_2 \in N$,

$$\varphi(n_1n_2) = (n_1n_2)M = n_1M \cdot n_2M = \varphi(n_1) \cdot \varphi(n_2).$$

The kernel of φ is

$$\ker(\varphi) = \{n \in N \mid \varphi(n) = M\} = \{n \in N \mid nM = M\} = \{n \in N \mid n \in M\} = N \cap M.$$

The image of φ is

$$\text{Im}(\varphi) = \{nM \mid n \in N\}.$$

Since every element of NM can be written as nm for $n \in N$ and $m \in M$, and since $\varphi(n) = nM$, we have that

$$\text{Im}(\varphi) = \{nM \mid n \in N\} = \{nmM \mid n \in N, m \in M\} = NM/M.$$

Therefore, $\text{Im}(\varphi) = NM/M$.

By the first isomorphism theorem, we have

$$N/\ker(\varphi) \cong \text{Im}(\varphi).$$

Substituting the kernel and image, we get

$$N/(N \cap M) \cong NM/M.$$

Therefore, we have shown that $NM/M \cong N/(N \cap M)$.

Appendix A

Conclusion

In conclusion, this report has navigated the intricate landscape of the Fundamental Theorem of Homomorphisms, highlighting its significance and applications within group theory. Beginning with the foundational elements of groups, subgroups, and cosets, we laid a robust foundation for understanding more complex concepts. The exploration of normal subgroups and quotient groups provided a deeper understanding of the structural hierarchy within groups.

Homomorphisms were then introduced as pivotal mappings that preserve group structure, leading to a detailed exposition of their kernels and images. The Fundamental Theorem of Homomorphisms was rigorously examined, emphasizing its role in connecting the kernel of a homomorphism to the structure of its corresponding quotient group. Through a series of examples and applications, we demonstrated how this theorem facilitates the analysis and classification of groups in diverse mathematical contexts.

By synthesizing these concepts and illustrating their practical implications, this report aimed to enhance the understanding and appreciation of the Fundamental Theorem of Homomorphisms, showcasing its profound impact on abstract algebra and its applications across various disciplines within mathematics.

Appendix B

References

- [1] I.N. Herstein, *Topics in Algebra*, 2nd, Wiley, 2006.
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